



Top Chemical GIFs

Chemistry can be absolutely beautiful at times. Here are some GIFs as proof: <http://dgit.in/1IK0rX4>



Microbes on Mars

Not alien life, but Curiosity might've taken microbes from Earth on to Mars <http://dgit.in/1gZdyYv>

The War Against Space Junk

The sky up above just can't be an unlimited litterbox

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Look up at the sky on a clear night. High up, away from the blinding glow of the metropolis you'll see a smattering of stars, colourful flickering planets and empty blackness. Over the centuries we've learnt that the sky is filled with so much more than what we can see. The many billions of planets, stars, meteors and other celestial bodies are the scattered remains of a once converged universe now spreading out in all directions. Everything we've ever seen in the heavens is touched by natural creation and is of ancient origins - well, almost everything.

Spatula From Space and Other Clutter

In 2006 during a routine testing mission near the International Space Station, astronaut Piers Sellers of the space shuttle Discovery accidentally lost a spatula. This spatula joined the other millions of man-made objects orbiting the Earth, with no purpose and no predictability. Along other artificially introduced debris of smaller size, such as metal shavings, paint flecks, frozen urine and other assorted items, it has become impossible to keep an accurate count.

But the larger fragments, those over one centimeter in size, have been successfully tracked out of necessity as they pose a direct threat to human space flight. The most recent count of space debris places



The Mir station's solar panels have already suffered major damage from space debris.

over 500,000 detectable fragments between the size of 1 cm and 10cm, and over 20,000 fragments about the size of a tennis ball floating around the Earth. These remnants orbit at speeds of over 28,000 kilometers per hour, which is 14 times the speed of sound, and are a constant threat to our orbital presences. The impact that even the smallest paint flecks can have on the equipment orbiting Earth can prove devastating. Already scores of these fragments are primed to hit communication satellites, defence satellites, the Interna-

tional Space Station and any other craft sent up to space if we don't detect them and adjust our flight plans.

The dramatic influence that space junk can have in the long run can easily be witnessed in the film *Gravity*. But to truly appreciate the relevance down here on Earth all you have to do is imagine suddenly losing all telecommunications facilities - no cell phones, no television, no GPS, no weather reports - all gone. Just because a spatula lost in 2006 came around and smashed violently into an orbiting com-